



Frequency-Domain Diffusion Models for Deepfake Analysis

Synthesizing Spectral Fingerprints with Diffusion Models

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⊕ Problem Definition

Motivation

Modern generative models can produce **highly realistic images** in the spatial domain, but these images often contain artifacts, in the frequency domain, which can act as **synthetic fingerprints** and can be exploited by forensic detectors to distinguish real images from generated ones.

Objective

Explore a novel generative paradigm: training Diffusion Models directly on the **frequency spectrum** rather than the standard RGB spatial domain. By modeling the true spectral distribution of real images, the goal is to produce spectra closer to real images and **reduce synthetic forensic fingerprints** while maintaining acceptable visual quality.





State Of Art

2020 – Frank et al.
*Frequency analysis of
GAN-generated images*

2023 – Corvi et al.
*Detection of
diffusion-generated images*

2020 – Frank et al.
*Frequency analysis of
GAN-generated images*

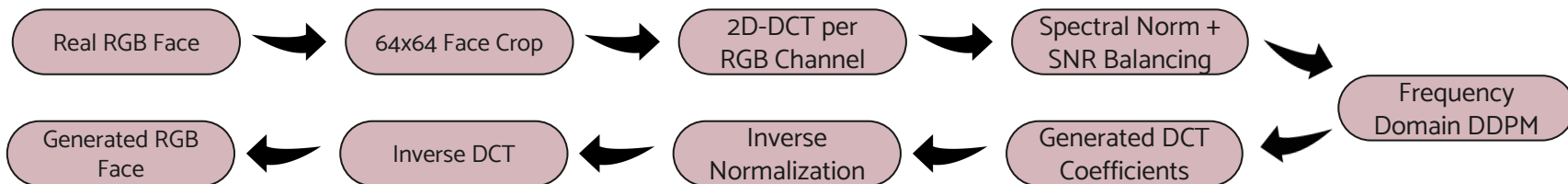
Combine these ideas from a forensic perspective: train an **unconditional diffusion model** directly on **DCT coefficients** and compare it against a conventional RGB diffusion model, with particular attention to whether the generated spectral fingerprints become closer to those of real images.





Proposed Method

Diffusion Directly in DCT Space:



RGB baseline:





Mathematical Foundations

1

DCT Transform: The image is converted to DCT coefficients using an orthonormal transform

$$D = CXC^T$$
$$X = C^TDC$$

2

Spectral normalization : Coefficients are normalized to reduce the large variance imbalance between low and high frequencies.

$$y_0 = \frac{D - \mu}{\sigma^{0.90}}$$

3

Forward diffusion : Standard DDPM noise is applied directly to this normalized spectral state.

$$y_t = \sqrt{\bar{\alpha}_t} y_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

4

v-prediction objective : U-Net is trained with v-prediction to learn the reverse process

$$v_t = \sqrt{\bar{\alpha}_t} \epsilon - \sqrt{1 - \bar{\alpha}_t} y_0$$
$$\mathcal{L} = \mathbb{E} \left[w_t \|v_\theta(y_t, t) - v_t\|_2^2 \right]$$





Dataset

The experiments use Celeb-DF, a face-forensics dataset containing both real and manipulated videos.

SPLIT	VIDEOS	FRAMES
<i>Training Real</i>	370	17.760
<i>Validation Real</i>	8	256
<i>Evaluation Real</i>	30	1.920
<i>Synthetic reference</i>	62	992

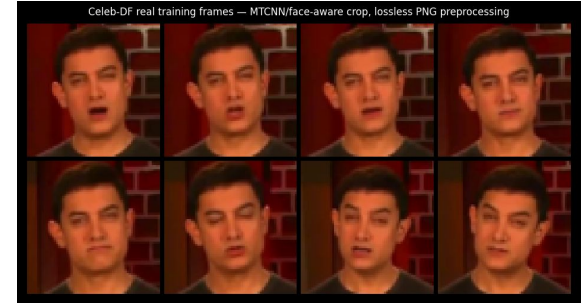


Figure 1. Examples of the real training face crops



Figure 2. Synthetic forensic-reference crops after the same preprocessing pipeline.





Experimental Setup

The main configuration is:

Network

Compact U-Net with the following changes:

- horizontal/vertical/radial frequency **coordinate map**;
- depth-wise 5x5 **frequency mixing**.

Diffusion

These are the main parameters used for the diffusion model:

- Resolution: 64 x 64
- Diffusion steps: 1000
- Schedule: Cosine
- Target: v-prediction
- Min-SNR γ : 5
- Self-conditioning: 50%





Training Setup

The training configuration uses:

Frequency learning rate:	1.5e-4
RGB learning rate:	2.0e-4
Weight decay:	2.0e-4
Gradient clipping:	1.0

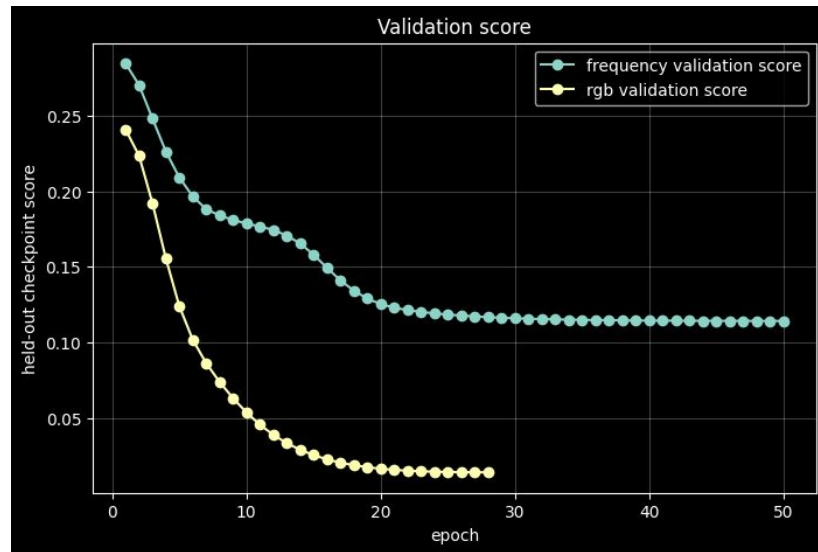


Figure 3. Validation-score trajectories observed during training for the frequency-domain model and the RGB baseline.






Evaluation

The project is evaluated in both image space and frequency space.

Visual Quality

FID
KID
Inception Score
Face Detection

Spectral Similarity

Mean Spectrum L1
Radial Profile L1
Band SWD
High Frequency
Energy
Peak/Grid Scores

Forensic Detectability

Grouped DCT AUC
Residual-FFT AUC
JPEG + Resize test
Video-grouped folds





Visual Results

Metric	RGB DDPM	Frequency DDPM
FID	116.31	162.15
KID	0.07612	0.12943
Inception Score	2.13 ± 0.16	1.93 ± 0.16



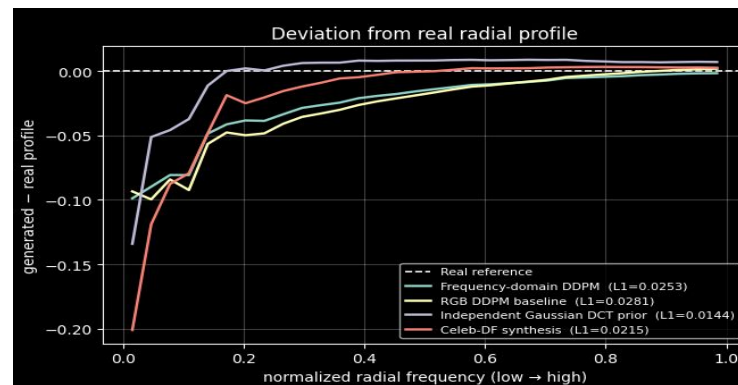
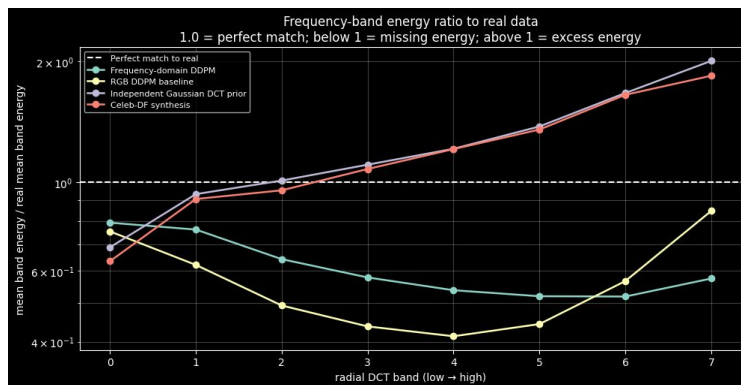
Figure 4. Generated faces from the frequency-domain DDPM (left) and RGB DDPM baseline (right). The RGB baseline is visually sharper and more coherent, while the frequency-domain model is evaluated primarily for its spectral behavior.





Spectral Results

Model	Mean-spectrum L1	Radial-profile L1	High-frequency energy / real	Band SWD	Grouped DCT AUC	Residual-FFT grouped AUC
RGB DDPM	0.01976	0.02820	0.456	0.736	0.969	0.957
Frequency DDPM	0.01764	0.02616	0.513	0.524	0.740	0.675





Conclusion

Goal Achieved

The experiments partially support the original hypothesis. The frequency-domain DDPM does not outperform the RGB baseline in **conventional image-quality metrics**. The RGB model achieves lower FID, lower KID and higher IS. However, the frequency-domain DDPM consistently performs better on several **spectral and forensic measurements**.

Limitations

- Images are generated only at **64 x 64 resolution**;
- The frequency model does **not match** the RGB baseline on **FID** or **KID**;
- The models are intentionally compact because of **limited GPU memory**;

